

EVIDENCE REVIEW • SEPTEMBER 2026

Full Endoscopic Spine Surgery

Current evidence and clinical practice
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Disclosures & Objectives

- Disclosure: No proprietary or industry relationships relevant to this content.
- Define the endoscopic spectrum: uniportal (FESS), biportal (UBE/BEDS), and endoscopic fusion (Endo-TLIF, K-LIF, ELIF).
- Appraise the 2024–2026 evidence: RCTs, meta-analyses, network meta-analyses, cost-effectiveness.
- Apply evidence to indications, patient selection, complications and learning curves.
- Identify gaps: what the evidence does NOT yet support.

Evolution of the Field

- 1997: Yeung — transforaminal endoscopic spine system (TESSYS).
- 2000s: Ruetten — full-endoscopic interlaminar discectomy and decompression (FESS).
- 2010s: Heo / Kim — unilateral biportal endoscopic surgery (UBE/BEDS): standard scope, separate working portal.
- 2020s: endoscopic fusion matures — Endo-TLIF, trans-Kambin LIF (K-LIF), ELIF; hybrid and navigation-assisted workflows.
- Scoping review 2026: 1,193 studies / 877,285 patients — publication volume rising steeply over the last decade.

Technique Taxonomy

- Transforaminal (TF) — through Kambin's triangle; best for foraminal/extraforaminal and migrated disc; spine-sparing.
- Interlaminar (IL) — via interlaminar window; central/paracentral disc, stenosis; cauda equina risk at L5-S1 with high iliac crest.
- Biportal (UBE/BEDS) — viewing + working portal; conventional instruments, larger decompressions; dominant growth area.
- Endoscopic fusion — posterior (Endo-TLIF/PLIF), lateral via Kambin's triangle (K-LIF/ELIF); instrumented, standalone or hybrid percutaneous fixation.
- Level distribution of the evidence base: lumbar >> cervical > thoracic.

The Evidence Pyramid — State of the Art (2026)

35.3%

of the literature is uncontrolled clinical effectiveness

31%

of studies report PROMs beyond 1 year

<4%

report cost or learning-curve outcomes

- Scoping review 2026: 35.3% of endoscopic spine literature is uncontrolled clinical effectiveness — quality is the field's weak link.
- Cost (3.6%) and learning-curve outcomes (4.4%) are the least-reported domains.
- But: multiple RCTs and network meta-analyses now exist for discectomy, decompression and fusion.
- PROM reporting: 90% capture baselines; only 31% report beyond 1 year — durability data still thin.
- Bottom line: equivalence to microsurgery is well-supported; superiority claims are mostly in secondary endpoints.

Global Spine J 2026 scoping review (PMC12975538). RCTs and NMA now exist for discectomy, decompression and fusion.

Lumbar Disc Herniation — Discectomy

- Systematic review + meta-analysis 2025 (87 studies, 3,238 patients): endoscopic lumbar discectomy (FELD/BELD) vs open & MIS.
- VAS leg pain significantly LOWER for endoscopic at 1 day AND 1 year.
- Recurrence and revision rates LOWER than open; dural tear and wound infection trending lower.
- Comparable to other MIS techniques (microendoscopic).

87 / 3,238

Studies and patients, 2025 meta-analysis

Endoscopic discectomy: VAS leg pain lower at 1 day and 1 year; recurrence and revision lower than open; ~85–95% meaningful improvement at 1–2 years — statistically equivalent to microsurgery.

J Clin Med 2025;14(19):6757. Liles 2025: opioid use 41.7% endoscopic vs 76.5% open.

Discectomy — Why Outcomes Differ

- Less muscle disruption / blood loss → lower VAS in the first postoperative days.
- Continuous irrigation + angled optics → direct visualization of the fragment and exiting root, less neural retraction.
- Day-case / short stay: many series report 0–1.3 days length of stay.
- Opioid consumption lower post biportal endoscopic surgery.
- Recurrence advantage likely technique-related: endoscopic discectomy is more fragment-selective, preserves annulus and posterior tension band.

Lumbar Stenosis — Decompression

- RCTs (biportal vs microscopic laminectomy): equivalent decompression adequacy and clinical outcomes at mid-term follow-up.
- Biportal: smaller incision, less blood loss, shorter hospital stay; longer operative time in learning phase.
- Safety profile comparable to open/minimally-invasive surgery for disc herniation and stenosis (Neurospine 2025 comparative safety analysis).
- Unilateral decompression with bilateral visualization (over-the-top) is a recognized endoscopic strength.
- No high-level evidence yet of superiority in PROs at ≥ 2 years — durability data pending.

Cost-Effectiveness — A Turning Point

- 2026 trial-based economic evaluation (Spine J): biportal endoscopic spine surgery (BESS) is cost-effective vs microscopic surgery (MSS) for lumbar degenerative disease.
- **Matched comparative series 2025 (thoracic):** open surgery associated with ~7× the cost of endoscopic surgery with comparable outcomes.
- **Drivers:** length of stay, opioid use, rehabilitation time — OR time is often longer for endoscopic.

~7×

Cost of open vs endoscopic surgery

Matched thoracic comparative series 2025:
OR time 342.8 vs 141.5 min, LOS 4.9 vs 0 days,
with comparable outcomes. First trial-based
economic evaluation (Spine J 2026) finds BESS
cost-effective vs microscopic surgery.

Spine J 2026 (S1529943026005851); Liles et al 2025.

Endoscopic Fusion — Where the Evidence Stands

- Network meta-analysis (Osseous Spine 2025): endoscopic posterolateral/interbody fusion, standalone LLIF, and comparators — no significant difference in pain, disability or adverse events; standalone LLIF best for operative time and LOS.
- BE-TLIF vs PLIF: similar clinical and radiologic outcomes; better early postoperative recovery with biportal (2025 comparative study).
- FE-KLIF (trans-Kambin) vs PLIF / MIS-TLIF: equivalent fusion rates; feasible and viable.
- Systematic review endoscopic vs MIS TLIF: no significant difference in pain scores, complications or fusion rates.
- Endoscopic fusion is real, but evidence is mostly retrospective comparative — no mature RCT program yet.

ELIF / Lateral Endoscopic Interbody Fusion

- First systematic review of ELIF (2025, NASS): 13 studies, 518 patients.
- Fusion: open pooled 98%, MIS 84%, endoscopic 100% (single series — interpret cautiously).
- Pooled VAS back: 6.49 → 1.66 (Δ 4.83 at ~13 months, endoscopic subgroup).
- Pooled ODI: 34.6 → 12.2 (endoscopic subgroup).
- Transient radiculopathy 9.5%, dural tear 0.5%.
- Requires favorable extraforaminal anatomy + intraoperative nerve monitoring.

Cervical & Thoracic Applications

- Systematic review full-endoscopic cervical surgery (Neurospine 2025): posterior PECF/PECD — VAS 7.8 → ~1.1; shorter stay, less blood loss; short-term safety acceptable.
- Single-level full-endoscopic posterior cervical foraminotomy: safe at 12 months (ORR 2025); evidence remains Level IV.
- Thoracic (JSS 2026 systematic review): trans-SAP and FESS for TDH/stenomyelopathy show large PROM gains, but Level III-IV only; 4 serious complications in a 41-patient meta-analysis — steep learning curve.
- Myelopathy: endoscopic decompression reported even for calcified central discs via transpedicular routes — expert-centre results, not generalizable.

Recurrent Disc Herniation

- 2025 systematic review (World Neurosurg): recurrent LDH reoperation — full-endoscopic revision offers alternative trajectories and improved visualization of scarred anatomy.
- TF approach avoids the posterior scar plane from prior open discectomy.
- PROMs after endoscopic revision are comparable to first-time endoscopic surgery in available series.
- Still no RCT for revision surgery — selection (central vs foraminal, scar, instability) drives outcome.

THE HONEST NUMBERS

Complication profile: comparable to microsurgery in rate — different in distribution, not absent.

- Dural tear: rare in disc surgery; higher in endoscopic stenosis work than in open series with senior hands.
- Nerve root injury / transient dysesthesia: characteristic of TF approach (exiting root) — most transient.
- Infection: discitis reported; continuous irrigation is protective when pressure is controlled.
- Irrigation-related risks: raised intracanal pressure, and rare tamponade / intracranial pressure events in cervical & thoracic work.
- Thoracic series: paralysis, CSF leak, infection reported — do not dilute indications during the learning curve.

Learning Curve — The Hidden Variable

30–50

supervised cases before independent practice

18.5 ± 9.2

cases per surgeon to LIF proficiency

80%

of operative-time gain arrives late in the cumulative series

- Meta-analysis of learning curves (2024): most rapid improvement in the first phase; roughly 30–50 supervised cases before independent TF discectomy.
- Large registry analysis: 80% of operative-time improvement arrives only after a long cumulative series.
- Full-endoscopic LIF proficiency cut-off $\approx 18.5 \pm 9.2$ cases per surgeon (2025 meta-analysis) — after prior endoscopic experience.
- Complication rates fall with the curve; conversion-to-open rate is the honest metric to track.
- Training: cadastre + simulation + proctored live cases; logbook-level case tracking is now standard of care in fellowship programmes.

Learning-curve meta-analysis 2024 (PMC11622208); J Clin Med 2025;14(24):8926; Int J Mol Sci 2024.

Technology Horizons 2025–26

- **Navigation- and robotics-assisted endoscopy:** portal planning for Kambin's triangle and pedicle-based fusion.
- 3D/4K/4K-3D fluorescence and IONM-integrated workflows for lateral endoscopic fusion.
- Unilateral biportal as the workhorse platform for decompression; endoscopic fusion expanding indications to low-grade spondylolisthesis.
- **Hybrid endoscopic-open surgery:** endoscopic decompression + percutaneous instrumentation.

2025–26

Where the technology is going

Navigation, robotics and AI-assisted workflows are converging on the endoscopic corridor — the bottleneck is evidence, not instruments.

Patient Selection — Evidence-Guided

- Best supported: unilesional radicular LDH (foraminal/extraforaminal → TF; central/paracentral → IL or biportal).
- Stenosis with radiculopathy ± neurogenic claudication: biportal decompression where surgeon volume supports it.
- Spondylolisthesis/instability requiring fusion: endoscopic fusion equivalent only in experienced hands; MIS-TLIF remains the benchmark.
- Red flags, progressive deficit, cauda equina: operate promptly — approach by your highest-volume technique.
- Central calcified thoracic discs and severe osteoporotic stenosis: expert centres, not day-surgery programmes.
- Contraindications: inability to tolerate positioning/irrigation; obliterated foramen without alternative corridor.

WHERE THE EVIDENCE STOPS

Claims the current evidence does NOT support.

- Superiority of endoscopic over microsurgery in validated long-term PROs — equivalence, not superiority.
- Endoscopic surgery for multilevel disease, deformity, or high-grade instability as routine.
- Cost savings claimed outside trial-based evaluations (OR time can erase LOS savings).
- Any specific implant/cage platform advantage — no comparative evidence.
- Wide expansion of indications during the learning curve — the complication data are the warning.

PRACTICAL TAKE-HOME POINTS

Five things to carry out of this room.

- 1. Endoscopic discectomy: equivalent-to-better pain and recurrence vs open; lowest tissue trauma; day-case pathway.
- 2. Biportal decompression: equal outcomes to microscopic with faster recovery; now cost-effective on trial-level health economics.
- 3. Endoscopic fusion: equivalent fusion and PROMs in comparative series — volume and mentorship decide safety.
- 4. Respect the learning curve — it is the strongest predictor of both outcomes and complications.
- 5. Long-term PROMs, cost, and training-outcome data remain the field's evidence gaps — contribute to registries.

Key References (2024–2026)

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